

PANAGIOTIS KIOUSIS

Email address: panagiotis.qcs@gmail.com **Nationality:** Greek
LinkedIn: https://www.linkedin.com/in/panagiotis-kiousis-135558264/
Website: https://panagiotisq.github.io/ **GitHub:** https://github.com/panagiotisq

ABOUT ME

Highly motivated Electrical and Computer Engineering student at the University of Patras.
Passionate about innovation and research. Currently at TU Graz, working in the Institute of Visual Computing (IVC) in the lab of Prof. Friedrich Fraundorfer.
Interests span:
Computer Vision (especially 3D vision), Machine Learning, computer Graphics, Robotics, Reinforcement Learning, 3D computational Geometry.
I also like combinatorics and electromagnetic fields (only student in a class of 330 to receive the top grade).
Passionate Problem-Solver | Analytical | Collaborative | Multidisciplinary

EDUCATION AND TRAINING

Master Thesis
Technical University of Graz, Graz, Austria [02/2026 – Current]
Thesis: "Learning Which Correspondences to Trust: Confidence-Weighted Event-Camera Localization in LiDAR Maps", under the supervision of prof. Friedrich Fraundorfer
This work led to a paper submitted to ICRA 2027, available on arXiv: arxiv.org/abs/2610.11967.

Thesis: hdl.handle.net/10889/32879.

MEng in Electrical and Computer Engineering
University of Patras, Patras, Greece [01/09/2021 – Current]
GPA: 8.49/10 (current)
Specialization: 3D computer vision and geometry; robotics and control. Selected courses: Computational Geometry, 3D Computer Vision, SLAM and State Estimation, Reinforcement Learning, Optimal Control, Machine Learning, Computer Graphics, Robotic Systems I, II, etc.

High School Diploma
1st Lyceum of Markopoulo [09/2018 – 06/2021]
Final grade: 19.4 / 20

HONOURS AND AWARDS

[11/07/2025] IEEE World Haptics Conference
Best Student Innovation Challenge winner at IEEE Worldhaptics 2025 Our project "Enhancing Emotional Communication in Shared Visuo-Haptic XR Environments through Vibrotactile Actuation" was selected as finalist and presented live at the conference in Suwon, South Korea (July 8–11) and won the TOP AWARD!
Link: https://www.vvr.ece.upatras.gr/news/winners-best-student-innovation-challenge-ieee-worldhaptics-2025/

[2021] Hellenic Statistical Authority
3rd prize in Probabilities and Statistics Contest

Hellenic Mathematical Society
2nd phase of Greek Mathematics Competition (3 consecutive years) 2019, 2020, 2021

[2019] ΕΕΦ
Finalist (16th position) of Greek Physics Competition

ΕΕΦ
Finalist of Greek Physics Competition 2020, 2021

[2021] Mathematical Student Festival
Award in Mathematical Riddles contest

[2022]
IEEE Extreme programming competition (finished in the top 24%)

[2020]
1st prize in TABLE TENNIS Tournament

WORK EXPERIENCE

Research Assistant **VVR (Visualization and Virtual Reality) Group, University of Patras**
[03/2025 – Current]
Under the supervision of prof. Konstantinos Moustakas:
Research Assistant in DIDYMOS- XR (DIgital, DYnaMic and respOnsible twinS for XR) , a three-year EU-funded project
– 3d computational geometry, computer vision
– pothole/bump detection project : pothole-detection: pothole/bump detection in urban environments
– change detection project : change-detection: change detection in 2d and 3d point clouds
Also working in various projects in the field of biomedical computer science, in collaboration with Prof. Athanasios Lourbopoulos.
– Built a stroke analyst with a Unet and SAM for clinical use: stroke-analyst-2, presented at the ESOC 2026 (European Stroke Organisation Conference)
– Built a nationwide emergency-response accessibility model using OSRM routing, OpenStreetMap data, and Python to visualize ambulance travel times to all hospitals in Greece.

PROJECTS

Computer Graphics / **Computational Geometry**
C++/OpenGL projects in computer graphics. shader programming, rendering, lighting algorithms, graphics simulation, geometry processing, Laplacian mesh processing, 3D terrain reconstruction from image.
– computer-graphics-semester-project
– 3d-terrain-from-image
– laplacian-mesh-processing

Robotics Various projects in the field of robotics. (forward and inverse kinematics, SE(3) trajectory design, dynamics simulator, velocity and force/torque controllers, Visual-Inertial odometry, domain randomization for sim-to-sim transfer in reinforcement learning) visual-inertial-odometry, domain-randomization-sim2sim

Operations Research / **Optimization** End-to-end fulfilment-network optimization: probabilistic demand forecasting, two-stage stochastic capacity planning (Benders decomposition), and online order assignment for a synthetic Iberian e-commerce fulfilment network. iberia-fulfilment-optimization

[10/2023 – 01/2024]
2024 IEEE AP-S Student Design Contest Participation in the 2024 IEEE AP-S Student Design Contest as a member of the RadarG(r)eeks team, where we designed a portable low-power radar system. Submission of the proposal: “On Body Radar System as Navigational Aid for the Visually Impaired”
Link: https://drive.google.com/file/d/18sjau8nJ0gN2QXGCEfydhYYUpKD83ZdV/view?usp=drive_link

Electronics Projects Design of a speaker, microphone. Design of a wireless energy transmition system.

Programming Expert in: Python, C++, C (built semester projects / reasearch work) Good knowledge: Java, Matlab and Website development: backend, frontend, database

AutoCAD Technical drawing : electrical and mechanical drawing

IEEE

IEEE activities
– Participation in the IEEE Extreme programming competition (finished in the top 24%)
– Participation in the IEEE student branch of the University of Patras.
– Seminars/Webinars of IEEE. (certificates in my LinkedIn profile)

LANGUAGE SKILLS

GREEK (mother tongue), ENGLISH (C2-Proficient), GERMAN (A1-Basic)

HOBBIES AND INTERESTS

Guitar, Mountain bike, Hiking, Table Tennis, Football, Travelling, Photography

DRIVING LICENCE

Driving Licence: B